A simulated crop row, as a rover sees it

photograph | depth frame | point cloud

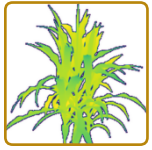
One model

The center plant out, and what it gives

recipe of the center plant | re-rendered plant | traits

sorghum







25 % of the center plant is hidden by its neighbors

CropCode

one encoder for photographs, depth, clouds and recipes; pretrained on single plants, then fine-tuned on crop rows to write the center plant

```
stem 1.12 m, 22 leaves
leaf start  len roll branch width
  1   0.19 0.27  171     20 0.091
  2   0.22 0.40  171     21 0.093
  3   0.26 0.54  314     21 0.093
  4   0.30 0.62  176     21 0.093
  5   0.34 0.66  320     21 0.100
     … 17 more leaves
```



Chamfer 19.6e-4 m² vs the true plant

height
1.12m (true 1.10m)
leaves
22 (true 21)
leaf length
0.75m (true 0.75m)
leaf width
9.9cm (true 10.4cm)

maize







22 % of the center plant is hidden by its neighbors

```
height 1.07 m, 13 leaves
leaf  dist  len   wid angle droop
  1  0.092 0.34 0.062    -7   -92
  2  0.091 0.36 0.063    -2   -91
  3  0.093 0.41 0.068    -4   -93
  4  0.092 0.44 0.064    13   -94
  5  0.093 0.44 0.063    -3   -95
     … 8 more leaves
```



Chamfer 2.6e-4 m² vs the true plant

height
1.07m (true 1.06m)
leaves
13 (true 13)
leaf length
0.36m (true 0.37m)
leaf angle
13° (true 13°)

Held-out test rows, seen broadside by the rover camera; nothing marks the center plant. The true plant is shown for comparison only.